📘 Full Stack Web Development Syllabus

Module 1: Frontend Development Basics

HTML 5

- Basic HTML
- Formatting
- Lists
- Tables
- Forms & Input
- Links, Frames
- Image, Audio, Video
- Style & Semantic
- Meta Info, Programming

CSS 3

- Basic (Foundations)
- Selectors
- Box Model
- Typography
- Backgrounds
- Basics & Modern Layouts
- Transitions & Animations
- Responsive Web Design
- Scroll & Overflow Styling
- CSS Variables & Functions
- CSS Multi-device & Printing
- Advanced Styling & Concepts

Bootstrap 5

- Introduction & Setup
- Core Concepts
- Typography
- Backgrounds & Borders
- Layout System
- Components (UI Elements)
- Forms & Tables

- Advanced Utilities
- JavaScript Powered Components
- Responsive Design Projects (Bootstrap)

Tailwind CSS

- Introduction & Setup
- Core Utilities
- Typography & Text Utilities
- Backgrounds & Borders
- Layout & Spacing
- Flexbox & Grid System
- Sizing & Positioning
- Transitions & Animations
- Advanced Utilities
- Responsive Design Projects (Tailwind)

JavaScript (ES6+ Features)

- JavaScript Basics (Syntax, Variables, Data Types, Operators)
- Control Statements (if/else, loops, switch)
- Functions (Normal, Arrow Functions, Parameterized, Recursive, return)
- Events & DOM Manipulation
- Asynchronous JS
- Error Handling
- JSON Handling
- Fetch API / Axios Concepts
- This Keyword, Scope, Hoisting
- Event Loop, Call Stack, Execution Context

Module 2: Modern Frontend

React JS

- Introduction to React (JSX & Virtual DOM)
- Core Concepts (Fundamentals)
- Styling in React
- React Hooks
- Forms & Controlled Components
- React Router
- State Management
- API Calls with Axios / Fetch

- React + Backend Integration
- Projects

Frontend Projects

- Portfolio Website
- To-Do App
- Weather App
- Ecommerce UI (Mini)

Module 3: Version Control & Deployment

Git & GitHub

- Introduction
- Git & GitHub - Basics
- Basic GitHub Repository Hosting

Module 4: Database (CRUD Operations)

Database - SQL (MySQL)

- Introduction of RDBMS
- Database, Table Creation
- Data Types
- CRUD Operations
- Constraints
- Operators
- Functions
- Aggregation & Grouping
- Simple Joins

Database - NoSQL (MongoDB)

- Introduction to NoSQL
- Collections & Documents
- Insert, Find, Update, Delete
- Query Operators

Module 5: Backend Development

Node JS

- Introduction & Setup
- NPM & Packages
- File System, Events
- Creating Simple Server

Express JS

- Routing
- Middleware
- REST APIs
- Error Handling
- Authentication & Authorization
- JWT / Session-based Auth
- Role-based Access Control
- Password Hashing (bcrypt)

Module 6: Integration & Projects

- Blog Application
- Ecommerce Store
- Social Media App